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Substance use stigma reduces trust and informational support: experimental evidence across risk and relationship contexts

Iván Flores Martínez^{1*}

Abstract

Background This study investigates the impact of addiction history on interpersonal trust and access to social support. Specifically, it explores whether people exhibit decreased trust toward individuals with addiction histories, explores the extent to which positive portrayals can mitigate such distrust, and evaluates whether those with addiction histories receive reduced social support from their closest social networks, including friends and family.

Methods Two complementary studies were conducted. Study 1 employed hypothetical scenarios to assess trust in both property-related and personal risk situations and examined whether counter-stereotypical information could reduce distrust. Study 2 used vignettes depicting individuals seeking assistance in differing relational contexts (friends versus family) to evaluate responses and types of support provided. Ordinary Least Squares (OLS) regression was applied to estimate the effects of experimental manipulations on trust and social support.

Results Participants exhibited significantly reduced trust toward individuals with a history of substance abuse across both scenarios. Providing counter-stereotypical information significantly reduced distrust in the property risk scenario, but had no measurable effect on trust in personal risk contexts. Regarding social support, individuals with substance use histories received less informational support when assistance was requested by family members.

Conclusions This research highlights the role of addiction stigma in shaping interpersonal dynamics, particularly in trust and social support. The findings suggest that targeted interventions addressing trust deficits, such as counter-stereotypical portrayals or trust-building programs, could play a role in reducing stigma and enhancing interpersonal relationships. Additionally, mentoring programs and improved access to informational resources may help bridge gaps in social support, especially where family support is limited. These insights have significant implications for rehabilitation programs, family support services, and community-based initiatives aimed at promoting successful recovery and reintegration.

Keywords Substance use stigma, Interpersonal trust, Social support, Counter-stereotypical information, Risk contexts, Relationship contexts, Informational support, Addiction history

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Introduction

Substance use disorders affect approximately 20.4 million adults in the United States alone, yet the stigmatization of individuals with a history of addiction remains a significant barrier to their recovery and reintegration into society [45]. This stigma manifests not only in institutional discrimination but in interpersonal dynamics that can shape recovery outcomes [12, 30]. Although research has documented the pervasive nature of addiction-related stigma and its detrimental effects on clinical outcomes and treatment participation [6, 9, 24, 40], a crucial gap remains in understanding how stigma affects interpersonal behaviors. In particular, the influence on trust and the willingness to offer support remains underexplored, even though both are essential for successful recovery [27, 46].

Recovery from substance use disorders frequently depends on the availability of robust social support networks and the ability to rebuild trusting relationships [21, 22]. However, stereotypes about individuals with addiction histories—particularly notions that they are inherently untrustworthy or dangerous—may undermine these resources [31]. Despite increasing recognition of this issue, three critical knowledge gaps persist. First, little is known about how addiction stigma affects interpersonal trust across risk contexts. Second, although stigma is understood to stem from deeply ingrained stereotypes, few studies have experimentally tested whether interventions such as providing positive portrayals of individuals in recovery can effectively mitigate its impact [35, 36]. Third, research has not adequately examined how stigma shapes the willingness of close support networks to provide different types of assistance to individuals with substance use histories.

To address these gaps, I present two complementary studies examining different facets of how substance use histories affects social relationships. Study 1 investigates how knowledge of someone's past substance use influences others' willingness to trust them in situations involving personal and property risks. It also explores whether providing counter-stereotypical information about an individual's current prosocial behaviors can help reduce initial distrust associated with their substance use history. Study 2 examines the complex dynamics of social support, focusing on how substance use history affects the willingness of friends and family members to provide various types of assistance. This dual approach allows for analysis of both the immediate impact of substance use stigma on interpersonal trust and its broader implications for accessing social support networks.

This research is grounded in two distinct but interconnected theoretical frameworks. Stigma theory emphasizes that individuals with substance abuse histories face both enacted stigma, defined as direct discrimination and

rejection, and perceived stigma, defined as the anticipation of negative treatment [9, 29]. These stigmas are reinforced by stereotypes that portray individuals with substance use histories as untrustworthy and dangerous [31]. Social support theory emphasizes the critical role of different support types (e.g. instrumental, informational, and emotional) in facilitating recovery and sustaining sobriety [1, 7]. Taken together, these frameworks illuminate how stigma can restrict access to critical social resources, creating a self-reinforcing cycle where diminished support further impedes recovery efforts.

Preliminary analysis reveals several key findings with important implications for harm reduction practice. Trust was significantly lower toward individuals with substance use histories in both personal and property risk contexts, with the effect being especially pronounced in property risk scenarios. Counter-stereotypical information effectively reduced distrust in property risk situations but had no measurable effect in personal risk contexts, suggesting that different domains of stigma may require distinct intervention strategies. Most notably, individuals with substance use histories received less informational support when seeking help from family members compared to friends. This counterintuitive finding challenges assumptions about kinship as a protective buffer against stigma. The selective withdrawal of informational support within families may represent a critical but overlooked barrier to recovery, as advice about resources, opportunities, and services is often essential for successful reintegration.

The findings have significant theoretical and practical implications. Theoretically, they advance understanding of the mechanisms through which stigma influences trust and social support, offering evidence for strategies to counteract its effects. Practically, the results can guide the development of targeted interventions including trust-building programs that address property risk concerns specifically, family education initiatives that emphasize the importance of informational support, and community stigma reduction campaigns that utilize counter-stereotypical narratives. These insights are particularly relevant for harm reduction services, rehabilitation programs, family support programs, and community efforts aimed at fostering successful recovery and reintegration.

The remainder of this paper is structured as follows: First, I review relevant literature on substance use stigma, interpersonal trust, and social support networks. Next, I present the methodology and results of Study 1, which examines how addiction stigma influences trust in different risk contexts and tests the effectiveness of counter-stereotypical information. I then describe Study 2, focusing on how stigma impacts access to different types of social support across relationship networks. Finally,

I discuss the implications of these findings for theory, practice, and future research.

Background

Stigma associated with substance use disorders is a complex, multidimensional phenomenon that operates at multiple levels of society. Link and Phelan [29] define stigma as arising when elements of labeling, stereotyping, separation, status loss, and discrimination co-occur within a power context. For individuals with substance use histories, this stigmatization process manifests in several distinct but interconnected forms. Public stigma is perhaps the most pervasive form, referring to the negative attitudes, beliefs, and discriminatory behaviors directed toward individuals with substance use disorders by the public [4, 36]. This form of stigma is sustained through widespread endorsement of negative stereotypes, which portray individuals with substance use disorders as dangerous, unpredictable, morally weak, and blameworthy for their condition [42, 50].

Empirical evidence consistently demonstrates that substance use disorders elicit stronger public stigma than other mental health conditions [9, 35, 42]. Public opinion surveys reveal that respondents express greater desires for social distance and stronger beliefs about personal responsibility when considering individuals with substance use disorders compared to those with depression or schizophrenia [38, 42]). This uniquely intense public stigma stems from perceptions of controllability and moral culpability [35]. Unlike many other health conditions, addiction is often viewed as a consequence of poor choices rather than a legitimate medical condition, leading to attributions of blame rather than compassion [9]. Media representations frequently reinforce these perspectives by portraying individuals with substance use disorders through crime-focused narratives or sensationalized imagery [31, 37], perpetuating views of substance use primarily through lenses of criminality and personal failure rather than as a treatable health condition.

Beyond public stigma, individuals with substance use histories also encounter structural stigma: discrimination embedded within institutional policies, practices, and procedures [18]. This manifests in healthcare systems through reduced insurance coverage for addiction treatment, exclusionary housing policies, employment discrimination, and legal penalties that create obstacles to recovery [30, 48]. Kennedy-Hendricks et al. [23] found that many healthcare providers hold negative views toward patients with substance use disorders, resulting in substandard care quality and reduced treatment participation. Similarly, barriers to employment and housing, essential components of stable recovery, often stem from structural stigma codified in background check requirements and rental policies [47].

The external stigma emanating from public and structural sources becomes internalized through processes of self-stigma and perceived stigma. Self-stigma occurs when individuals accept and apply negative societal attitudes toward themselves, leading to diminished self-esteem, reduced self-efficacy, and lower likelihood of seeking treatment [10, 11, 32]. Perceived stigma, defined as the anticipation of discrimination regardless of whether it occurs, can prompt individuals to withdraw from social interactions and avoid treatment settings in ways that may feel protective but are ultimately harmful [5, 28]. Matthews et al. [33] documented how these internalized forms of stigma correlate strongly with depression, anxiety, and substance use relapse among individuals in recovery.

Stigma and interpersonal trust

Trust represents a critical element in human relationships, defined as the willingness to be vulnerable to another based on positive expectations of their intentions or behaviors [41]. For individuals with substance use histories, rebuilding trust constitutes a fundamental challenge in recovery, as substance use often damages interpersonal relationships through behaviors that violate expectations of trustworthiness [21, 22]. However, perceptions of untrustworthiness may persist well beyond active substance use and into stable recovery, reflecting the enduring nature of addiction stigma rather than current behavioral patterns [49].

Research suggests that judgments about trustworthiness draw upon both cognitive and affective elements, with cognitive trust based on evaluations of competence and reliability, while affective trust emerges from emotional bonds and perceptions of goodwill [34]. Both dimensions may be compromised by substance use stigma. Cognitively, stereotypes about addiction may lead others to question the reliability, competence, and consistency of individuals with substance use histories regardless of their current behavior [50]. Affectively, negative emotional responses including fear, anger, and moral disgust may inhibit the formation of emotional connections necessary for affective trust [17].

Importantly, trust judgments appear context-dependent, with risk type potentially moderating the effect of substance use stigma on trust decisions. Although some research has examined general trust attributions toward stigmatized groups [44], few studies have investigated how substance use history affects trust across distinct risk domains such as personal safety versus property. Addressing this gap is critical, as recovery outcomes often depend on access to both personal support networks and material resources controlled by others [20, 27].

Countering stigma through individuating information

Stereotypes function as cognitive shortcuts, allowing people to make quick judgments based on category membership rather than individual characteristics [14, 15]. However, research on stereotype reduction suggests that providing individuating information (i.e. details about specific behaviors and attributes) can prompt more nuanced evaluations that override categorical judgments [25]. This perspective-taking process has shown promise in reducing stigma toward various marginalized groups, including people with mental illness [2, 36].

For substance use disorders specifically, McGinty et al. [35] found that media portrayals depicting successful recovery narratives improved public attitudes compared to portrayals focusing only on active substance use. Similarly, controlled experiments demonstrate that providing information about an individual's recovery journey and positive depictions of their current prosocial behaviors can attenuate negative evaluations based on substance use history [30]. However, these studies have primarily measured general attitudes rather than specific behavioral intentions such as trust, leaving open questions about whether individuating information affects willingness to engage in trust-based interactions with individuals who have substance use histories.

Social support in addiction recovery

Social support represents a multidimensional construct encompassing various forms of assistance provided through interpersonal relationships. Cohen & Wills [7] classify social support into four primary categories: emotional support (expressions of empathy, care, and reassurance), instrumental support (tangible aid and services), informational support (advice and guidance), and appraisal support (feedback relevant to self-evaluation). Each support type serves distinct functions in the recovery process, with their relative importance varying across recovery stages and individual circumstances [46].

Extensive research demonstrates the importance of social support in predicting positive recovery outcomes. Longitudinal studies consistently find that individuals with stronger support networks maintain longer periods of abstinence and report greater quality of life in recovery [26, 27, 32]. However, accessing these vital support resources often proves challenging for individuals with substance use histories, as stigma may constrain both the availability and quality of support across relationship networks [21].

Support availability may vary systematically across relationship types, as family members, friends, romantic partners, and professional contacts may respond differently to substance use disclosure [46]. Family relationships, characterized by strong emotional bonds but also histories of addiction-related conflict, can produce

complex support dynamics with high emotional support but potentially limited instrumental or informational resources [13]. Peer relationships, particularly with non-using friends, may offer alternative support patterns more focused on social integration but constrained by stigma-based social distancing [3, 46]. Despite these distinctions, little research has systematically examined how substance use stigma differentially affects access to specific types of support across these relationship domains.

Implications for current research

The literature reveals clear gaps in understanding how substance use stigma operates within interpersonal contexts to influence both trust judgments and support provision. While substantial research has documented the prevalence and broader effects of public, structural, self, and perceived stigma, less attention has focused on the specific interpersonal mechanisms through which stigma undermines recovery capital. Additionally, while emerging evidence suggests that individuating information may reduce general stigmatizing attitudes, it remains unclear whether such interventions can specifically increase trust-based behaviors toward individuals with substance use histories.

The current research addresses gaps in understanding how substance use stigma affects interpersonal trust and access to social support. Two complementary studies were conducted. Study 1 examines how substance use history influences trust across distinct risk contexts and tests whether counter-stereotypical information can mitigate this effect. It is hypothesized that individuals with a history of substance use will be trusted less than those without such a history in both property-related and personal risk scenarios, and that counter-stereotypical information is expected to reduce this distrust across contexts. Study 2 investigates how substance use history shapes access to different types of social support within close relationship networks. It is hypothesized that individuals with substance use histories will receive less social support overall, with family members more likely than friends to provide assistance. By examining these interpersonal dimensions of stigma, the research aims to advance theoretical understanding of stigma mechanisms and inform practical approaches to supporting recovery through social resource mobilization.

Study 1: Substance use history and trust

Design and participants

Study 1 employed a factorial design incorporating both between-subjects and within-subjects elements. Substance use history (present vs. absent) and risk context (personal vs. property risk) were manipulated as between-subjects factors, with participants assigned to distinct groups based on these conditions. Trust scores

were measured within subjects across two time periods: before and after exposure to counter-stereotypical information presented in the form of a newsletter. Participants were recruited through Amazon's Mechanical Turk (MTurk) and compensated \$2.50 USD for completing the experiment. The original sample included 500 individuals; after excluding participants who failed attention checks, the final sample consisted of 470 U.S. residents.¹ Participants were predominantly male (60%) and white (82%), with an average age of 23.04 years ($SD=10.18$). Most held an associate's degree (68%), reported being unmarried (73%), and had household incomes between \$50,000 and \$74,999 annually.

Materials and procedure

Participants were randomly assigned to read one of four vignettes describing a hypothetical classmate. The vignettes varied across two experimental conditions: whether the classmate had a history of substance use (treatment condition) or not (control condition), and whether the interaction involved personal risk or property risk. In the personal risk scenario, the classmate offered to visit the respondent's home to work on a class project while the respondent's roommates were away. In the property risk scenario, the classmate offered to watch the respondent's home while they were away on vacation. After reading the vignette, participants completed the initial trust measures.

Following these initial assessments, all participants received additional information about the classmate in the form of a newsletter from their college. This newsletter stated that the individual had received an award from a public service organization, providing counter-stereotypical information designed to challenge potential negative perceptions in scenarios involving substance abuse history. After receiving this information, participants completed the trust measures again, along with two additional items evaluating perceived change in the classmate and whether the new information had altered their attitudes.

Trust measures

Five survey items derived from previous trust research [8, 44] were used to operationalize trust. These items measured different dimensions relevant to the vignette scenarios: worry or fear about the described individual, perceptions of competence, willingness to be vulnerable, and expectations of positive outcomes from the interaction. Reliability analyses yielded acceptable Cronbach's alpha values across all conditions ($\alpha=0.70-0.77$),

Table 1 Cronbach Alpha Coefficients for Trust in Different Scenarios and Periods

Survey items	α	$sd(\alpha)$
Trust index – personal risk (period 1)	0.70	0.72
Trust index – personal risk (period 2)	0.74	0.76
Trust index – property risk (period 1)	0.73	0.74
Trust index – property risk (period 2)	0.77	0.77

Period 1 refers to responses collected before participants received the newsletter

Table 2 Baseline covariate balance by treatment group

Variable	Control (mean)	Treatment (mean)	Control (sd)	Treatment (sd)
Age	22.82	23.25	9.42	10.89
Political Party	1.62	1.56	0.77	0.77
Gender	1.39	1.56	0.49	0.50
Latino	1.69	1.70	0.46	0.46
Race	1.24	1.21	0.72	0.61
Education	6.76	6.65	1.15	1.15
Income	8.20	8.03	2.52	2.56

supporting the creation of a composite trust index by averaging responses across the five items (Table 1).²

Randomization check

Participants were randomly assigned to experimental conditions using Qualtrics' built-in randomizer, which implemented a simple randomization procedure. This method ensured that each participant had an equal probability of being assigned to any condition. Demographic summary statistics confirmed the effectiveness of the randomization, with treatment and control groups showing similar distributions across age, political affiliation, gender, ethnicity, race, education, and income variables. No significant differences were found between groups on any demographic variables, and no randomization failures occurred (Table 2).

Control variables

As control variables, demographic information was collected from respondents, including political affiliation (Democrat, Republican, Independent, Other), age (in years), gender (Male, Female, Other), marital status (Never married, Separated, Divorced, Widowed, Married, Living with a partner), number of children under 18 living at home, race/ethnicity (White, Black, American Indian, Asian, Other), highest level of education completed (ranging from less than 6th grade to a graduate degree), and total household income for 2021 (reported in categorical brackets from less than \$5,000 to \$300,000 or more).

¹The attention check questions encompassed a mathematical equation at the survey's outset and an inquiry about the project's due date described in the vignette. Participants were required to select the correct option from four choices provided.

²The Cronbach alpha test is a measure of internal consistency and scale reliability. It measures how closely related a set of items are as a group.

Manipulation checks

A formal manipulation check was not included due to concerns about demand characteristics, as directly asking participants whether they noticed the substance use manipulation could artificially inflate attention to this factor. However, participants were asked whether the newsletter influenced their perception of the classmate's character or habits (see Appendix). While not designed as a manipulation check, this item offers some insight into how the intervention may have shaped responses.

The newsletter was administered to all participants after exposure to one of four vignette conditions, which varied whether the classmate was described as having a substance use history and the risk context. Across both the personal and property risk scenarios, a greater proportion of participants in the substance use condition responded "Yes" compared to those in the control condition (92 vs. 80 in the personal risk scenario; 96 vs. 87 in the property risk scenario). These patterns suggest that the intervention was noticed and may have influenced participants' perceptions, despite the absence of a formal manipulation check.

Analytical approach

The analysis employed two regression models to evaluate the impact of addiction history and counter-stereotypical information on trust:

Model 1 (Baseline Trust): Using trust scores from the initial period as the outcome variable to demonstrate baseline differences in trust between treatment and control groups.

Model 2 (Trust Change): Using the difference in trust scores across periods as the outcome variable, capturing changes associated with the counter-stereotypical information intervention.

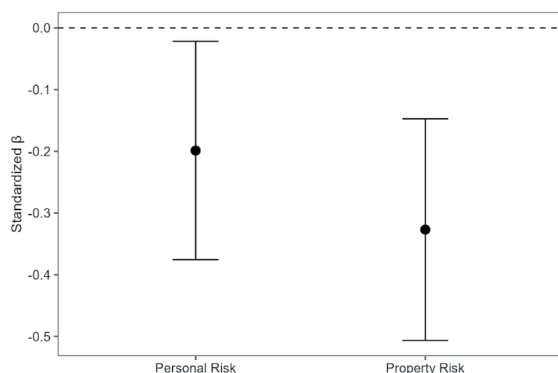


Fig. 1 Standardized Coefficient Plot for Substance Use History by Risk Scenario Before Newsletter. Note: Error Bars Represent 95% Confidence Intervals

Both models controlled for demographic factors such as age, gender, marital status, and income. The formal regression equation used was:

$$Y_i = \beta_0 + \beta_1 X_1 + Z' \beta_x + \varepsilon$$

where: Y_i represents the outcome variable. In Model 1: Trust score in the initial period (before intervention). In Model 2: Difference in trust scores (post-intervention minus pre-intervention). X_1 indicates group assignment (1=treatment/substance abuse history, 0=control). Z' represents a vector of demographic controls (e.g., age, gender, marital status, income), with β_x as their corresponding coefficients. ε accounts for unexplained variability in trust scores.

Separate analyses were conducted for personal risk and property risk scenarios to identify differences in trust dynamics across contexts.

Results

Initial differences across groups

Figure 1 presents a coefficient plot illustrating the negative impact of substance use history on trust across both personal and property harm scenarios. In the personal risk scenario, the standardized regression coefficient for addiction history was $\beta = -0.199$ ($p = 0.0278$), indicating a modest reduction in trust. In the property risk scenario, the standardized coefficient was $\beta = -0.327$ ($p = 0.00039$), reflecting a more substantial negative effect. These findings suggest that substance use history consistently reduces trust, with the effect being more pronounced when material possessions are involved. The results highlight the enduring stigma faced by individuals with substance use histories and emphasize the importance of addressing trust deficits in contexts involving perceived risk.

Effect of counter-stereotypical information (intervention)

To measure the effect of providing counter-stereotypical information, the difference in trust scores (calculated as post-intervention minus pre-intervention) for each participant was computed and regressed on the treatment variable (substance use history). The analysis controlled for demographic factors, including age, gender, marital status, race, education, income, religiosity, ideology, and political affiliation (Fig. 2).

Figure 2 presents a coefficient plot illustrating the context-dependent effects of the intervention on trust. In the personal harm scenario, the standardized regression coefficient for substance use history was $\beta = 0.137$ ($p = 0.397$), indicating a slight non-significant increase in trust among participants assigned to the individual with a history of substance use. In contrast, the property harm scenario produced a larger and statistically significant effect

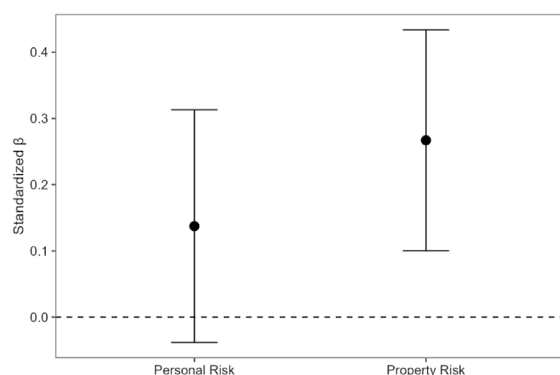


Fig. 2 Standardized Coefficient for Substance Use History by Risk Scenario After Newsletter. Note Error Bars Represent 95% Confidence Intervals

($\beta = 0.267$, $p = 0.0018$), reflecting a meaningful increase in trust following the intervention. These findings suggest that counter-stereotypical information effectively reduced distrust in the property harm context but had no measurable impact in the personal harm scenario. The results highlight the context-dependent nature of stigma reduction interventions and highlight the importance of tailoring interventions to distinct domains of perceived risk.

Study 2: substance use history and social support provision

Design and participants

Study 2 employed a 2×2 between-subjects factorial design manipulating substance use history (present vs. absent) and relationship type (family member vs. friend). Participants were recruited through Connect by Cloud Research and compensated \$0.80 for their participation. An a priori power analysis was conducted to estimate the minimum sample size required to detect a medium effect ($d = 0.5$) with 95% statistical power at an alpha level of 0.05 (two-tailed). Based on this estimate, the target sample size was set at 250 to accommodate potential exclusions and maintain adequate statistical power.

Out of the initial pool of 250 participants, 241 completed the study. After applying attention checks, the final analytical sample consisted of 231 participants. The attention checks included a basic mathematical equation presented at the beginning of the survey and a question about the project's due date described in the vignette. Participants were required to select the correct option from four multiple-choice options. The sample was predominantly male ($n = 125$) and largely composed of White individuals ($n = 195$), followed by Black participants ($n = 28$), American Indian participants ($n = 3$), and 10 individuals identifying with other minority groups, including Samoan, Guamanian, Hawaiian, Japanese, Korean, Asian Indian, Vietnamese, Filipino, and Chinese. In terms of age distribution, the sample was balanced across categories, with 55 participants aged 18–29 years, 66 aged 30–44 years, 65 aged

Table 3 Cronbach alpha coefficients for social support indexes

Survey items	α	No. of items
Social Support Index	0.824	12
Instrumental Support Index	0.723	4
Informational Support Index	0.644	4
Emotional Support Index	0.627	4

45–59 years, and 55 aged 60 years or older. According to the Cloud Research platform, the sample was representative of the U.S. population based on 2020 census data.

Materials and procedure

Participants read a hypothetical vignette describing a situation in which they recently spent time with someone they knew well who confided they were going through difficulties and requested assistance. The vignettes varied along to two experimental conditions: whether the individual had a history of substance abuse and was in rehabilitation (treatment) or not (control), and whether the individual was described as a family member (cousin) or a childhood friend. After reading the vignette, participants completed survey items measuring their willingness to provide different types of social support to the described individual.

Social support measures

Twelve survey items adapted from prior research [1] assessed participants' willingness to provide social support. These items measured three dimensions of social support:

1. Instrumental Support (4 items): Tangible assistance such as lending money, helping find a job or housing, and making social introductions ($\alpha = 0.723$).
2. Informational Support (4 items): Providing advice, counseling about life goals, sharing information about opportunities, and providing information about available social services ($\alpha = 0.644$).
3. Emotional Support (4 items): Offering a listening ear, physical comfort (e.g., a hug), engaging in fun activities, and setting aside frustrations when interacting with the individual ($\alpha = 0.627$).

Participants rated their likelihood of providing each form of support on a 5-point scale ranging from “not at all likely” (1) to “extremely likely” (5). An overall social support index was created by averaging responses across all twelve items ($\alpha = 0.824$), and separate indices were created for each support dimension (Table 3).

Analytical approach

Multiple OLS regression models were used to analyze the effects of substance use history, relationship type, and

their interaction on willingness to provide social support. The basic model specification was:

$$Y_i = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + Z' \beta_x + \varepsilon$$

where: Y_i represents the outcome variable, either overall social support or a specific support dimension (instrumental, informational, or emotional). X_1 indicates group assignment (1=treatment/substance abuse history, 0=control). X_2 indicates relationship type (1=cousin, 0=friend). Z' represents a vector of demographic controls (minority status, age, gender, income, and education). ε accounts for unexplained variability.

Separate models were estimated for overall social support and each support dimension. This approach allowed for examination of whether the effects of substance use history differed based on relationship context and varied across different types of support.

Results

Overall social support

Participants demonstrated greater willingness to provide social support to family members compared to friends ($\beta = 0.462$, $p = 0.3039$), suggesting that kinship positively influences support provision (Fig. 3).

While substance use history alone did not significantly influence overall support provision, its interaction with family relationship yielded a significant and negative effect ($\beta = -0.544$, $p = 0.038$). This standardized coefficient indicates that participants were less willing to offer social support to family members with substance use histories compared to friends with similar backgrounds.

Support dimensions

Analysis of specific support dimensions using standardized regression coefficients revealed distinct patterns:

1. **Instrumental support:** Minority status was associated with increased willingness to provide instrumental support ($\beta = 0.436$, $p = 0.0177$). Neither substance use history nor relationship type significantly affected instrumental support provision.
2. **Informational support:** Participants provided more informational support to family members than to friends ($\beta = 0.267$, $p = 0.0107$). However, the negative interaction between family status and substance use history ($\beta = -0.345$, $p = 0.0182$) indicates that participants were significantly less willing to provide informational support to family members with substance use histories. This pattern is further illustrated in the marginal effects plot (Fig. 4), which shows that when substance use history is considered, predicted informational support decreases for family members while slightly increases for friends. These findings suggest that substance use history may erode support more sharply within familial relationships than among peers.
3. **Emotional support:** The experimental conditions did not significantly affect the provision of emotional support, indicating that participants' willingness to provide emotional comfort remained consistent regardless of substance use history or relationship type.

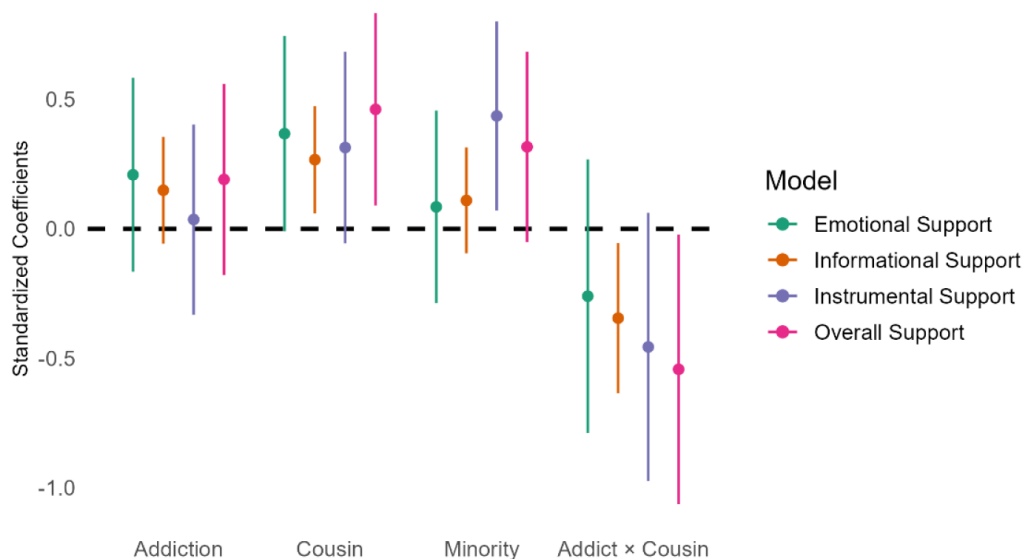


Fig. 3 Standardized Coefficient Plot from Regression Analyses Predicting Social Support. Note Error Bars Represent 95% Confidence Intervals

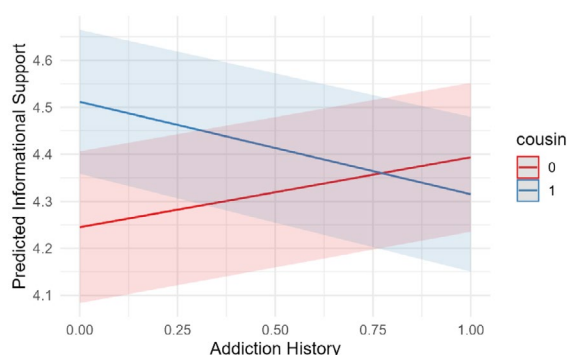


Fig. 4 Standardized Coefficient Plot from Regression Analyses Predicting Informational Social Support. *Note* Shaded Regions Represent 95% Confidence Intervals Around Each Coefficient Estimate

Demographic factors including age, income, education, and gender did not significantly predict social support provision across any dimension.

Robustness check

A Bonferroni correction was applied to the interaction term (*Substance Use History* × *Cousin*) across the models examining instrumental, informational, and emotional support, in order to account for multiple comparisons. With three tests conducted, the adjusted significance threshold was set at $p < 0.0167$. After correction, none of the interaction effects reached statistical significance (Instrumental Support: $p = 0.238$; Informational Support: $p = 0.055$; Emotional Support: $p = 0.976$). Although these adjusted p-values warrant caution in interpreting statistical significance, the direction of the effects remain consistent with theoretical expectations, lending support to the broader interpretation of the findings.

Discussion

This research examined how substance use history influences interpersonal trust and access to social support: two critical resources for successful recovery. The findings reveal distinct patterns in how stigma operates across different risk and relationship contexts with important implications for both theory and practice.

Differential trust impacts across risk contexts

Study 1 demonstrated that substance use history negatively influenced trust in both personal and property risk scenarios, with the effect notably stronger in the property risk context. Specifically, the standardized effect size in the property risk scenario was approximately 64% greater than in the personal risk scenario, indicating a more pronounced trust deficit. This finding aligns with prior research suggesting that addiction stigma activates specific stereotypes of untrustworthiness [31, 50], and it extends this work by emphasizing the context-dependent nature of these effects. The stronger impact

in property-related situations suggests that stereotypes linking substance use with theft or financial irresponsibility may be especially salient in public perceptions. This interpretation is consistent with research highlighting increased concern about material risk when individuals interact with members of stigmatized groups [31, 39, 44].

Effectiveness of counter-stereotypical information

The intervention using counter-stereotypical information yielded context-dependent results. In the property risk scenario, participants exposed to positive information about individuals with a substance use history demonstrated significantly higher trust ($\beta = 0.267$, $p = 0.0018$), suggesting that the intervention successfully mitigated stigma in this domain. This finding supports theories of stereotype reduction through individuating information [25], indicating that highlighting prosocial achievements, such as public service or personal growth, can override categorical judgments about material trustworthiness. In contrast, the same intervention did not significantly alter trust in the personal risk scenario ($\beta = 0.137$, $p = 0.397$). This discrepancy suggests that certain dimensions of stigma, particularly those involving perceived threats to personal safety, appear more resistant to information-based interventions. Dual-process models of stigma [17] provide a useful framework for interpreting this result: while cognitive stereotypes about competence or responsibility may be modifiable through positive exemplars, affective reactions rooted in fear or moral judgment often persist despite corrective information. In contexts where interpersonal risk is perceived as high, trust may be governed more by emotional intuitions than by rational reassessment.

Taken together, these findings underscore the importance of tailoring stigma reduction strategies to specific domains of perceived risk. Interventions that rely on counter-stereotypical messaging may be more effective in addressing material concerns than in shifting deeply held affective responses related to personal safety.

Complex dynamics of social support

Study 2 examined how substance use history interacts with relationship type to shape willingness to provide social support. While substance use history alone did not significantly reduce overall support, its interaction with relationship type revealed important nuances. Participants expressed greater willingness to support family members than friends; however, this preference reversed when the individual had a substance use history. In those cases, participants were less willing to provide support to family members than to friends, challenging the assumption that kinship bonds buffer against stigma. Instead, these findings suggest that family relationships may sometimes amplify stigma-related withdrawal of support.

Dimension-specific analyses further clarified these dynamics. The interaction effect was strongest for informational support, where participants were significantly less willing to offer advice, guidance, or resource-related help to family members with substance use histories ($\beta = -0.345$, $p = 0.0182$). This suggests that stigma may selectively restrict access to critical recovery resources within familial networks. The mechanism underlying this pattern remains unclear from the current data, though several explanations warrant consideration. Family members may withhold informational support due to doubts that advice would be followed, concerns about enabling, or lingering resentment from past betrayals during active substance use. Alternatively, families may view providing information as interference or may fear being blamed if recommendations fail to produce successful outcomes.

In contrast, instrumental and emotional support were less affected, indicating that some forms of support may be more resilient to stigma. This pattern aligns with prior research showing that substance use stigma can strain family relationships in targeted ways while leaving other support dimensions intact [13]. The relative stability of emotional support across conditions is particularly noteworthy, suggesting that existing emotional bonds may serve as a foundation for rebuilding damaged relationships even when other forms of support are compromised.

The exploratory finding that racial minority participants reported greater willingness to provide instrumental support ($\beta = 0.436$, $p = 0.0177$) merits brief mention. While not part of the original analytic plan and therefore preliminary, it may reflect heightened empathy among individuals from marginalized backgrounds who are more attuned to the social costs of stigma [2, 19, 43]. Future research should examine whether shared experiences with discrimination or cultural values around communal support contribute to these patterns.

Theoretical implications

These findings advance stigma theory in several ways. First, they demonstrate that addiction stigma affects interpersonal processes in context-specific rather than uniform ways, with distinct patterns across risk domains and relationship types. This supports conceptualizations of stigma as multidimensional rather than monolithic [29] and underscores the importance of domain-specific measurement in stigma research.

Second, the variable effectiveness of counter-stereotypical information across contexts advances our understanding of when and how stereotype reduction interventions succeed. The significant reduction in property-related trust deficits suggests that interventions emphasizing achievements and responsibility can effectively target specific dimensions of stigma. In contrast,

the persistence of trust deficits in personal risk scenarios highlights the limitations of such approaches, particularly when affective concerns, such as fear or moral judgment, are involved. These findings support both the contact hypothesis and exemplar-based theories by showing that exposure to positive representations can reshape attitudes, especially when tailored to address context-specific biases.

Third, the finding that family relationships may amplify rather than buffer stigma effects challenges the assumption that kinship inherently protects against social exclusion. Instead, the results suggest that stigma interacts with relationship history in complex ways, activating distinct concerns in familial versus peer contexts. This contributes to emerging theories that examine how stigma operates within close relationship networks, and not just between strangers, and calls for greater attention to relational dynamics in stigma research [33].

Practical implications

These findings offer several actionable insights for improving recovery support and reducing stigma. First, the successful reduction of property-related trust deficits through counter-stereotypical information suggests that recognition programs highlighting the achievements of individuals in recovery may effectively reduce specific stigma dimensions. Organizations could develop initiatives that publicize recovery success stories, particularly those emphasizing responsibility and reliability, to counteract stereotypes related to material trustworthiness.

Second, the persistence of trust deficits in personal risk scenarios indicates that certain domains require more intensive or alternative intervention strategies. Information-based approaches alone may be insufficient; trust-building programs should incorporate structured opportunities for positive contact in supervised settings to address deeper affective concerns tied to personal safety. Third, the reduced provision of informational support to family members with substance use histories highlights a specific gap that support programs should address. Family education initiatives should focus on overcoming information-sharing barriers by offering structured opportunities for relatives to learn about recovery resources and develop skills for providing effective informational support. Programs should also address underlying concerns such as fear of enabling, doubts about receptiveness to advice, or unresolved resentment from past experiences.

Fourth, the relative stability of emotional support across conditions suggests that existing emotional bonds may serve as a foundation for rebuilding damaged relationships. Support programs can leverage these preserved connections while working to strengthen other dimensions of support that are more vulnerable

to stigma. To move beyond general stigma reduction, harm reduction and recovery support programs should incorporate targeted strategies that address relational dynamics and perceived trustworthiness. Specifically, trust-building interventions should include:

- **Relational repair modules:** Facilitate structured conversations between individuals with substance use histories and their family members to rebuild trust, clarify expectations, and address past relational ruptures.
- **Perspective-taking exercises:** Help support providers understand the lived experiences of those with substance use histories, reducing moral judgment and increasing empathy.
- **Communication skills training:** Equip both parties with tools to express needs, boundaries, and concerns in ways that promote mutual respect and reduce defensiveness.
- **Support mapping tools:** Encourage participants to identify who they trust and feel supported by and explore gaps or barriers in those relationships.
- **Cultural sensitivity components:** Tailor interventions to account for cultural norms around family obligation, shame, and substance use, especially in communities where familial roles are tightly defined.

Limitations and future directions

Several limitations warrant consideration. First, both studies employed hypothetical scenarios rather than observing actual behavior, raising questions about generalizability to real-world interactions. Social desirability bias may also have led participants to over-report willingness to help someone with a substance use history. However, the detection of significant stigma effects despite this potential bias suggests that actual behavioral differences may be even more pronounced. Future research could strengthen ecological validity by incorporating behavioral measures of trust and support provision in naturalistic settings.

Second, the cross-sectional design limits causal inference, particularly regarding how relationships between individuals with substance use histories and their support networks evolve over time. Longitudinal studies tracking trust and support patterns throughout recovery would provide valuable insights into these dynamics.

Third, concerns about sampling and external validity merit attention. Both studies relied on online research panels which are known to introduce demographic biases. Participants from these panels tend to be younger, more highly educated, less politically diverse, and more familiar with experimental paradigms than the general population [16]. The predominantly young, white, and

male composition of our samples further restricts generalizability. As such, the findings primarily reflect the perceptions of this demographic, and generalization to broader or more diverse populations should be made cautiously. Future research should strive for more representative sampling, including greater diversity in age, race, socioeconomic status, and geographic location.

Fourth, Bonferroni corrections were applied to account for multiple comparisons in the analysis of interaction effects across support dimensions. After adjustment, none of the interaction terms remained statistically significant. While this suggests caution in interpreting these effects as definitively established, the directional consistency and theoretical relevance of the findings remain noteworthy. Future research should consider larger sample sizes to better detect subtle interaction effects and employ preregistration to distinguish confirmatory from exploratory analyses.

Fifth, although a small proportion of participants were excluded based on attention check failures (6% in Study 1; 7.6% in Study 2), these exclusions were based on objective criteria designed to ensure data quality. Given the randomized design and relatively low exclusion rates, the risk of systematic selection bias is minimal. Nonetheless, excluded participants may have differed in meaningful ways (e.g., attentiveness, comprehension), and this limitation should be considered when interpreting the findings.

Finally, future research should examine specific mechanisms underlying the reduced informational support from families, explore whether findings generalize across different substances and recovery stages, investigate how stigma effects vary across diverse cultural contexts, and test whether the observed patterns hold in real-world interactions rather than hypothetical scenarios. Additionally, research examining how individuals with substance use histories perceive and respond to differential support patterns could provide important insights for developing more effective interventions.

Conclusion

This research demonstrates that stigma associated with substance use history operates through distinct interpersonal mechanisms that vary systematically across contexts. Individuals with substance use histories face significant trust deficits, particularly in situations involving property risk, though counter-stereotypical information can effectively mitigate these concerns in material domains. More surprisingly, family members, traditionally viewed as primary sources of recovery support, may selectively withhold informational support, creating barriers to accessing critical resources needed for successful reintegration. These findings challenge assumptions about stigma as uniformly negative and kinship as

inherently protective, revealing instead a complex landscape where relationship history, risk perceptions, and support types interact in ways that can either facilitate or hinder recovery.

For harm reduction practice, these results underscore the need for multifaceted interventions tailored to specific domains of stigma. Recognition programs highlighting achievements may reduce property-related trust concerns, while structured contact experiences and relational repair work may be necessary to address deeper affective barriers. Family support programs should specifically target informational support provision, helping relatives overcome hesitations about sharing advice and resources. By understanding how stigma operates differently across contexts and relationships, practitioners and policymakers can design more precise and effective strategies to dismantle barriers to recovery and promote successful community reintegration for individuals with substance use histories.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12954-025-01334-5>.

Supplementary Material 1

Author contributions

Iván Flores Martínez designed the study, oversaw data collection, conducted the statistical analysis, interpreted the results, wrote the initial draft of the manuscript, revised and edited the manuscript drafts, prepared all figures, and reviewed and approved the final manuscript. Finally, he updated the manuscript with the reviewers' comments.

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Data availability

The data supporting the findings of these studies are available from the author upon request.

Declarations

Ethics approval

Study 1 was approved by the Office of Research Protections and Integrity at UNC Charlotte under protocol number IRBIS-17-0070. The study title is *Omnibus Experiments in Information Processing and Public Opinion, M-Turk Studies*, and approval was granted on March 22, 2022. Study 2 was approved by the Office of Research Protections and Integrity at UNC Charlotte under protocol number IRB-23-0815. The study title is *Breaking the Stigma: Uncovering Factors Behind Mistrust of Those with Substance Abuse History*, and approval was granted on May 9, 2023.

Informed consent

All participants provided informed consent prior to participation in both studies.

Competing interests

The authors declare no competing interests.

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